

Growing a Sympoietic Art Organism: An Experimental Framework for Tracking the Micro-Dynamics of Cultural Evolution

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Empirical data recording how cultural artefacts, motifs, and lineages form in real time are of great benefit and interest in the study of cultural evolution. In this work, we introduce an experimental design (nicknamed “*Sympoietic Art Organism*” in a nod to Haraway[1]) that attempts to richly capture the bottom-up emergence of a shared cultural heritage in a small, controlled setting, as well as the influences at play throughout its evolution. Our approach calls for a small community of artists to engage in active collaborative artmaking over several months to build up a shared creative commons. After an initial “seeding phase” round of art contributions to the commons data repository, participants are asked to contribute at least five new artefacts to the commons every two weeks, over a series of “co-creative cycles”, and with the requirements that each shared piece engage with one or more pre-existing artworks (that is, artworks from earlier cycles) and that we record these influences between works to a shared ledger.

We, the ten co-authors of this study, took part in our own experiment over three months, growing a dataset of 411 original artworks, and a network of 725 verified relationships between them. Our experiment yielded a complete history of the creation of cultural artefacts (artworks) by each participant, and of all acknowledged influences from other pieces, allowing us to reconstruct the citation network of contributed artworks. By enforcing synchronous updates within each co-creative cycle such that participants only engaged with artworks from earlier stages, we were able to model the network’s growth as a time series.

Our results are well suited for multi-modal analysis. Quantitatively, we observed the structural evolution of the citation network from disconnected nodes into a consolidated core as the system matured. We found the network’s structural dynamics were consistent rich-get-richer feedback, showing that preferential attachment emerged endogenously around specific hub artworks. Qualitatively, the artworks provide a basis for tracing a kind of semantic phylogeny through visual inspection and comparison of contributed artworks, to see how participants adapted, remixed, or subverted specific aesthetic motifs across generations.

We propose this experiment as a proof of concept for a scalable ethnographic tool in network science. This approach provides a transparent environment for investigating the mechanisms of cultural inheritance.

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[1] D. J. Haraway, *Staying with the Trouble: Making Kin in the Chthulucene*. Duke University Press, 2016.

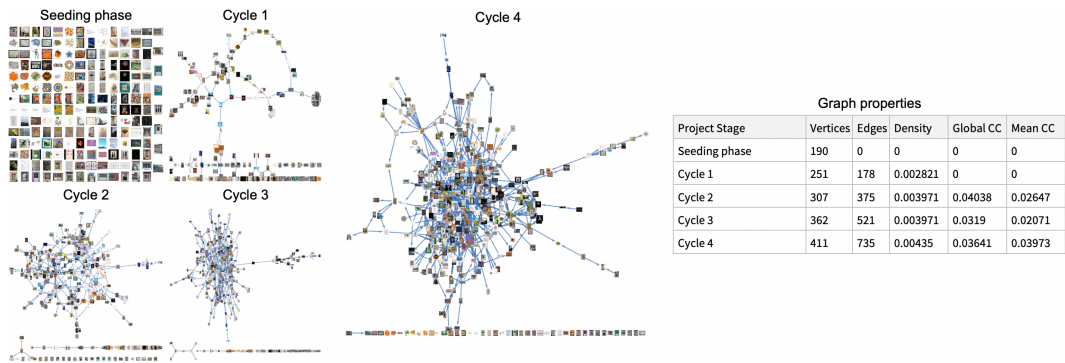


Figure 1: **Evolution of the art citation network.** Left: evolution of the citation network structure across generative cycles. Right: Summary table of graph properties at each stage of the experiment.